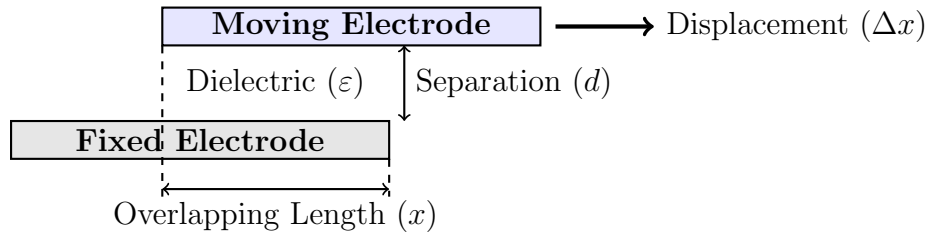


**B.E. Electrical Engineering Second Year Second
Semester Examination 2025**
ELECTRICAL INSTRUMENTATION PART -I

Examination Solutions

1.(a) In capacitive level gauges for conducting liquids, the conducting liquid serves as one of the electrodes.

Below is the 2D cross-sectional diagram of the variable area capacitive displacement transducer. The diagram illustrates the overlapping region which dictates the capacitance.



Note: Both plates have a constant width w going into the plane of the paper.

Mathematical Derivation

For a parallel plate capacitor, the fundamental capacitance C is given by:

$$C = \frac{\epsilon A}{d} \quad (1)$$

Where:

- ϵ is the absolute permittivity of the dielectric medium between the plates.
- A is the effective overlapping area of the plates.

- d is the constant separation distance between the plates.

Assuming the plates are rectangular with a constant width w and an initial overlapping length x , the overlapping area is $A = w \cdot x$. Substituting this into the equation gives the initial capacitance:

$$C = \frac{\varepsilon wx}{d} \quad (2)$$

When the movable electrode is displaced longitudinally by a distance Δx , the overlapping length changes, causing the overlapping area to change by $\Delta A = w \cdot \Delta x$. This results in a proportional change in capacitance ΔC :

$$\Delta C = \frac{\varepsilon w \Delta x}{d} \quad (3)$$

The **Sensitivity** (S) of a displacement transducer is defined as the change in output per unit change in input. Here, it is the change in capacitance divided by the change in displacement:

$$S = \frac{\Delta C}{\Delta x} = \frac{\varepsilon w}{d} \quad (4)$$

This derivation proves that for a variable area capacitive transducer, the sensitivity is completely linear and independent of the displacement x .

Numerical Solution

Based on the provided problem statement, we have the following parameters:

Given Data:

- Initial Capacitance, $C_0 = 100$ pF (*Note: This is extraneous information and is not required to calculate sensitivity*).
- Change in displacement, $\Delta x = 10$ mm
- Change in capacitance, $\Delta C = 10$ pF

Calculation:

Using the fundamental definition of sensitivity derived above:

$$S = \frac{\Delta C}{\Delta x} \quad (5)$$

Substituting the given values into the formula:

$$S = \frac{10 \text{ pF}}{10 \text{ mm}} \quad (6)$$

$$S = 1 \text{ pF/mm} \quad (7)$$

Final Answer:

The sensitivity of the variable area capacitive transducer is **1 pF/mm**.

Question 1.(b)

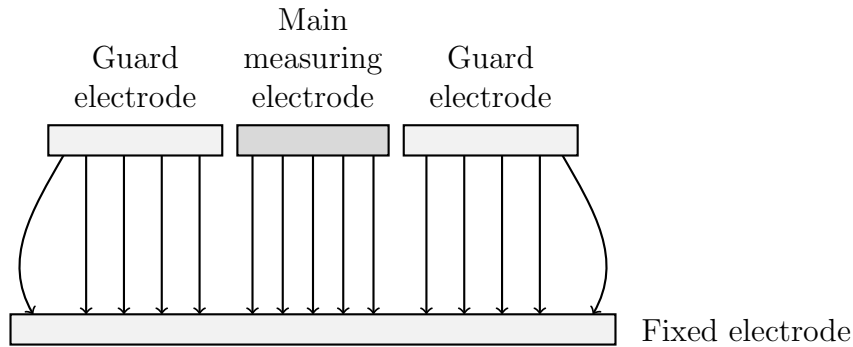
”The fringing effect of electric field lines in capacitive sensors can be minimized by opting for a push-pull (differential) configuration.”

Verdict: This statement is **Incorrect**.

Correction: The fringing effect is minimized by using a **Guard Electrode**. A **push-pull (differential) configuration** is used to eliminate **non-linearity** and increase sensitivity.

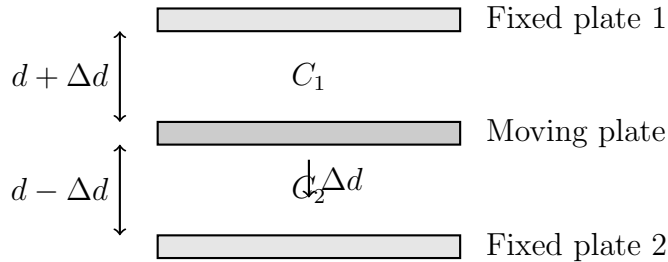
1. Minimizing Fringing Effect (Guard Electrode)

Fringing causes electric field lines to bulge at the edges, making the capacitance-displacement relation non-linear. To minimize this, a **Guard Electrode** is placed around the main measuring electrode and kept at the same electrical potential. This forces the electric field lines under the active measuring electrode to remain perfectly straight. Fringing only occurs at the outer edges of the guard, which is excluded from the measurement circuit.



2. Purpose of Push-Pull Configuration (Eliminating Non-Linearity)

A push-pull capacitive sensor uses a moving central plate between two fixed plates.



If the central plate moves downwards by Δd , the new capacitances are:

$$C_1 = \frac{\epsilon A}{d + \Delta d} \quad \text{and} \quad C_2 = \frac{\epsilon A}{d - \Delta d} \quad (8)$$

When placed in a bridge circuit, the output voltage V_{out} is proportional to $\frac{C_2 - C_1}{C_2 + C_1}$. Substituting the values:

$$V_{out} \propto \frac{\frac{\epsilon A}{d - \Delta d} - \frac{\epsilon A}{d + \Delta d}}{\frac{\epsilon A}{d - \Delta d} + \frac{\epsilon A}{d + \Delta d}} = \frac{\Delta d}{d} \quad (9)$$

Conclusion: Because $V_{out} \propto \Delta d$, the output is perfectly linear. Thus, a push-pull configuration is utilized strictly to eliminate the non-linearity of a single capacitor, not to reduce fringing.

Question 1.(c)

”In LVDTs, rectifier circuits may be deployed for getting an output replicating the pattern of variation of the displacement (being sensed) with time.”

Verdict: This statement is **Incorrect**.

Correction: Ordinary rectifier circuits **cannot** replicate the true pattern of displacement because they lose directional (phase) information. A **Phase-Sensitive Demodulator (PSD)** or Synchronous Demodulator must be deployed.

In-Depth Justification

1. The Nature of the LVDT Output Signal:

According to BJB Sir's notes, the LVDT output is an Amplitude Modulated (AM) suppressed-carrier signal. If the excitation is $e_i(t) = A \cos(\omega_c t)$, the output voltage is given by:

$$e_0(t) = -K_0 \cdot x_i(t) \cdot e_i(t) = -AK_0 \cdot x_i(t) \cos(\omega_c t) \quad (10)$$

Where K_0 is the sensitivity and $x_i(t)$ is the core displacement. When the core crosses the null point, $x_i(t)$ changes sign, causing a strict 180° **phase reversal** in the high-frequency carrier signal $e_0(t)$.

2. The Failure of Ordinary Rectifiers:

An ordinary full-wave rectifier only extracts the absolute magnitude of the signal envelope. It acts mathematically as an absolute value function:

$$V_{rectifier}(t) \propto |x_i(t)| \quad (11)$$

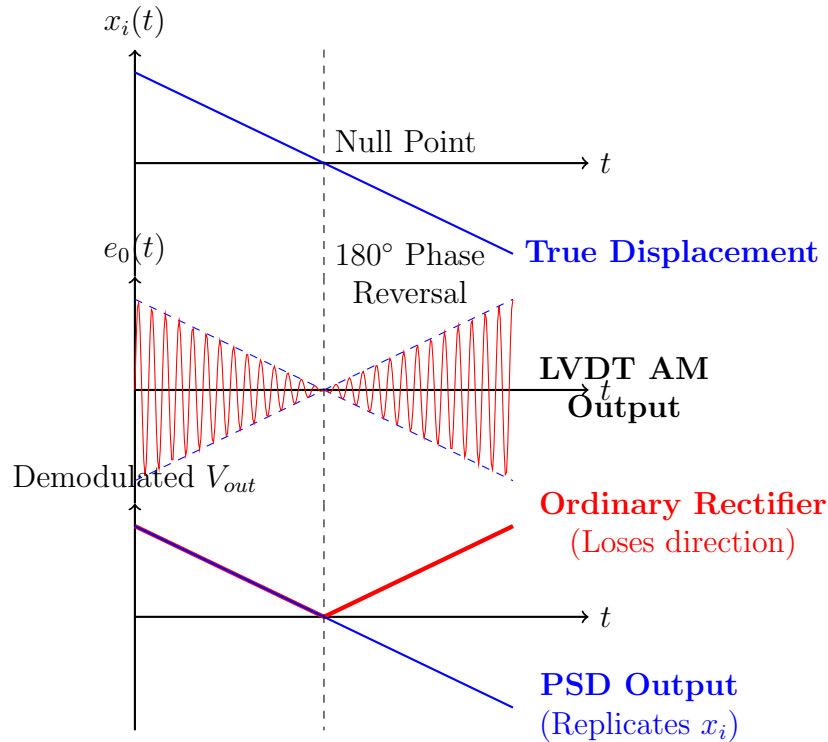
Because of this, an ordinary rectifier will convert a negative displacement into a positive output voltage. It cannot differentiate whether the core is at $+x$ or $-x$ distance from the null point.

3. The Solution (Phase-Sensitive Demodulation):

To replicate the true pattern of $x_i(t)$, a Phase-Sensitive Demodulator (PSD) is required. A PSD uses the primary excitation voltage $e_i(t)$ as a reference to detect the phase reversal. By multiplying the LVDT output with the reference and passing it through a low-pass filter, the PSD successfully recovers both the magnitude and the algebraic sign of the displacement.

Graphical Proof (From Lecture 4 & 5b)

Consider a case where the core travels at a constant speed towards the null point, crosses it, and continues moving.



Question 1.(d)

”Piezoelectric transducers are well-suited for measuring static mechanical force.”

Verdict: This statement is **Incorrect**.

Correction: Piezoelectric transducers are active devices well-suited **only for dynamic (time-varying)** measurements. They are strictly unsuitable for measuring static (constant) mechanical forces.

In-Depth Justification & Derivation

1. Charge Generation Principle:

According to BJB Sir’s notes, when a mechanical force F is applied to a piezoelectric crystal, it generates an electric charge q proportional to the applied force:

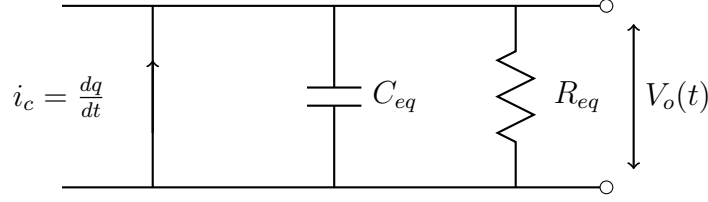
$$q = d \cdot F \quad (12)$$

Where d is the charge sensitivity constant (C/N).

2. Equivalent Electrical Circuit:

The crystal acts as a charge generator. However, the piezoelectric material and the

attached measuring instrument inherently possess an equivalent capacitance (C_{eq}) and a finite equivalent leakage resistance (R_{eq}). The generated charge slowly leaks away through this resistance.



3. Mathematical Derivation (Transfer Function):

Taking the time derivative of the charge equation gives the equivalent current source: $i_c = \frac{dq}{dt} = d \cdot \frac{dF}{dt}$. Applying Kirchhoff's Current Law (KCL) to the circuit:

$$i_c = i_C + i_R \quad (13)$$

$$d \frac{dF(t)}{dt} = C_{eq} \frac{dV_o(t)}{dt} + \frac{V_o(t)}{R_{eq}} \quad (14)$$

Taking the Laplace transform (assuming zero initial conditions):

$$d \cdot sF(s) = C_{eq}sV_o(s) + \frac{1}{R_{eq}}V_o(s) \quad (15)$$

Rearranging for the transfer function gives a **High-Pass Filter** characteristic:

$$\frac{V_o(s)}{F(s)} = \frac{d \cdot R_{eq} \cdot s}{1 + sC_{eq}R_{eq}} = \frac{K\tau s}{1 + \tau s} \quad (16)$$

Where $\tau = R_{eq}C_{eq}$ is the time constant and $K = d/C_{eq}$ is the voltage sensitivity.

4. Response to a Static Force:

A static (constant) force applied at $t = 0$ acts as a step input: $F(t) = F_0 \implies F(s) = \frac{F_0}{s}$. Substituting this into our equation:

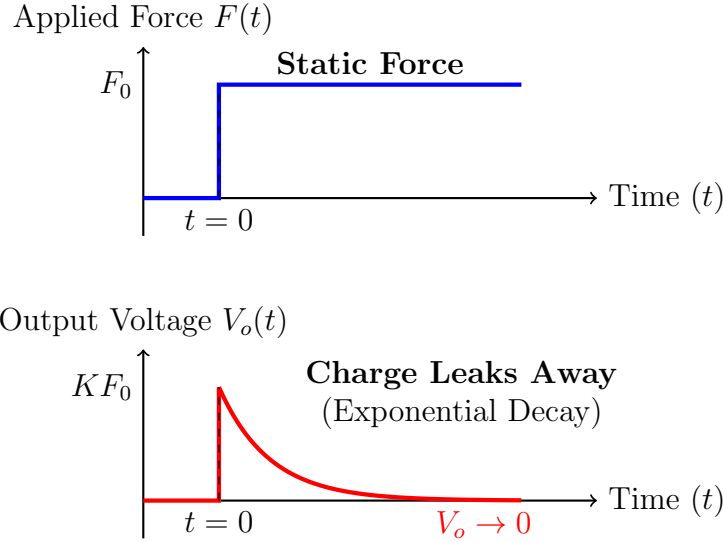
$$V_o(s) = \left(\frac{K\tau s}{1 + \tau s} \right) \frac{F_0}{s} = \frac{K\tau F_0}{1 + \tau s} = \frac{KF_0}{s + \frac{1}{\tau}} \quad (17)$$

Taking the inverse Laplace transform yields the time-domain output voltage:

$$V_o(t) = KF_0 e^{-\frac{t}{\tau}} \quad (18)$$

5. Final Conclusion & Graphical Proof:

The equation $V_o(t) = KF_0 e^{-t/\tau}$ mathematically proves that when a static force is applied, the output voltage initially spikes, but then **decays exponentially to zero** as the generated charge leaks away through R_{eq} . Since the steady-state output is zero, piezoelectric sensors cannot measure static forces.



Question 1.(e)

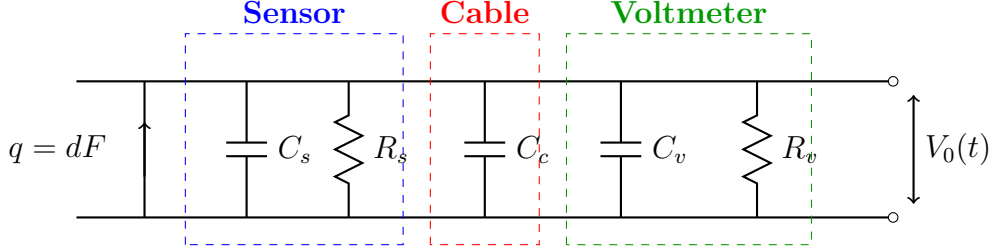
Problem Statement: Connections from a piezoelectric sensor are taken to a voltmeter via a coaxial cable 20 m long. The sensor material has a 'd' constant of 150 pC/N. Capacitance of the sensor is 100 pF and the leakage resistance is $10^4 \text{ M}\Omega$. Input capacitance of the voltmeter is 20 pF and its input resistance is 200 M Ω . The variety of coaxial cable used has a capacitance of 40 pF/m. Calculate the voltage indicated by the voltmeter 2 seconds after a sudden force of 10 N is applied.

1. Extracted Data

- **Sensor Charge Sensitivity (d):** 150 pC/N = $150 \times 10^{-12} \text{ C/N}$
- **Sensor Capacitance (C_s):** 100 pF = $100 \times 10^{-12} \text{ F}$
- **Sensor Leakage Resistance (R_s):** $10^4 \text{ M}\Omega = 10,000 \times 10^6 \Omega$
- **Cable Capacitance (C_c):** 20 m $\times$ 40 pF/m = 800 pF = $800 \times 10^{-12} \text{ F}$
- **Voltmeter Capacitance (C_v):** 20 pF = $20 \times 10^{-12} \text{ F}$
- **Voltmeter Resistance (R_v):** 200 M $\Omega = 200 \times 10^6 \Omega$
- **Applied Force (F_0):** 10 N (Step input at $t = 0$)
- **Time (t):** 2 seconds

2. Equivalent Circuit Formulation

According to BJB Sir's piezoelectric transducer notes, the sensor, the connecting cable, and the readout instrument (voltmeter) all act in **parallel**.



Total Equivalent Capacitance (C_{eq}):

Because the components are in parallel, capacitances add up:

$$\begin{aligned} C_{eq} &= C_s + C_c + C_v \\ &= 100 \text{ pF} + 800 \text{ pF} + 20 \text{ pF} \\ &= 920 \text{ pF} = 920 \times 10^{-12} \text{ F} \end{aligned}$$

Total Equivalent Resistance (R_{eq}):

Resistances combine in parallel:

$$\begin{aligned} R_{eq} &= \frac{R_s \cdot R_v}{R_s + R_v} \\ &= \frac{(10,000 \text{ M}\Omega) \cdot (200 \text{ M}\Omega)}{10,000 \text{ M}\Omega + 200 \text{ M}\Omega} \\ &= \frac{2,000,000}{10,200} \text{ M}\Omega \approx 196.078 \text{ M}\Omega = 196.078 \times 10^6 \Omega \end{aligned}$$

3. Time Constant & Initial Voltage Calculation

The **Time Constant** (τ) of the equivalent circuit governs the charge leakage rate:

$$\begin{aligned} \tau &= R_{eq} \cdot C_{eq} \\ &= (196.078 \times 10^6 \Omega) \times (920 \times 10^{-12} \text{ F}) \\ &= 180.392 \times 10^{-3} \text{ seconds} \approx \mathbf{0.1804 \text{ s}} \end{aligned}$$

The **Total Generated Charge** (q) from the sudden step force is:

$$\begin{aligned} q &= d \cdot F_0 \\ &= (150 \times 10^{-12} \text{ C/N}) \times 10 \text{ N} \\ &= 1500 \times 10^{-12} \text{ C} \end{aligned}$$

The **Initial Peak Output Voltage** (V_{peak}) at $t = 0^+$ is:

$$\begin{aligned} V_{peak} &= \frac{q}{C_{eq}} \\ &= \frac{1500 \times 10^{-12} \text{ C}}{920 \times 10^{-12} \text{ F}} \approx \mathbf{1.6304 \text{ V}} \end{aligned}$$

4. Final Transient Response

When a sudden static force is applied to a piezoelectric transducer, the resulting output voltage decays exponentially due to the finite equivalent leakage resistance. The governing equation is:

$$V_0(t) = V_{peak} \cdot e^{-t/\tau} \quad (19)$$

Substituting our calculated values to find the voltage at $t = 2$ seconds:

$$\begin{aligned} V_0(2) &= 1.6304 \cdot e^{-\frac{2}{0.180392}} \\ &= 1.6304 \cdot e^{-11.087} \\ &= 1.6304 \cdot (1.531 \times 10^{-5}) \\ &\approx 2.496 \times 10^{-5} \text{ V} \end{aligned}$$

Final Answer: The voltage indicated by the voltmeter 2 seconds after the sudden force is applied is approximately **24.96 μV** .

Question 2.(a)

”In transit-time ultrasonic flowmeters in the wetted sensor variety, the differential transit time of the ultrasound is affected by the temperature of the liquid.”

Verdict: This statement is **Correct** (True).

In-Depth Justification & Derivation

1. System Setup & Variables:

Consider a pipe where a liquid is flowing with a velocity v . Two wetted ultrasonic transducers (A and B) are mounted across from each other, separated by an acoustic path length L , at an angle θ relative to the direction of the fluid flow.

- v = Flow velocity of the liquid
- c = Speed of sound in the liquid
- L = Acoustic path length between sensors
- θ = Angle of the ultrasonic beam relative to fluid flow

2. Transit Time Equations:

The effective velocity of the ultrasound wave is altered by the velocity of the moving fluid.

- **Downstream Transit Time (t_1):** When the pulse travels *with* the flow (from A to B), the fluid’s velocity component adds to the speed of sound:

$$t_1 = \frac{L}{c + v \cos \theta} \quad (20)$$

- **Upstream Transit Time (t_2):** When the pulse travels *against* the flow (from B to A), the fluid's velocity component subtracts from the speed of sound:

$$t_2 = \frac{L}{c - v \cos \theta} \quad (21)$$

3. Differential Transit Time (Δt):

The flowmeter operates by measuring the difference between these two transit times:

$$\Delta t = t_2 - t_1 \quad (22)$$

Substituting the transit time equations:

$$\begin{aligned} \Delta t &= \frac{L}{c - v \cos \theta} - \frac{L}{c + v \cos \theta} \\ &= \frac{L(c + v \cos \theta) - L(c - v \cos \theta)}{c^2 - v^2 \cos^2 \theta} \\ &= \frac{2Lv \cos \theta}{c^2 - v^2 \cos^2 \theta} \end{aligned}$$

In almost all practical applications, the speed of sound in liquids (e.g., $c \approx 1500$ m/s for water) is vastly greater than the fluid velocity (v). Therefore, $c^2 \gg v^2 \cos^2 \theta$, allowing us to simplify the denominator:

$$\Delta t \approx \frac{2Lv \cos \theta}{c^2} \quad (23)$$

4. The Role of Temperature:

The speed of sound in a liquid (c) is a physical property dictated by the liquid's bulk modulus (K) and density (ρ), given by the relation $c = \sqrt{K/\rho}$. Both K and ρ are **highly dependent on temperature**.

Because c^2 sits in the denominator of our final expression for Δt :

- If the temperature of the liquid changes, the speed of sound c changes.
- Consequently, the raw differential transit time Δt will change, even if the actual flow velocity v remains perfectly constant.

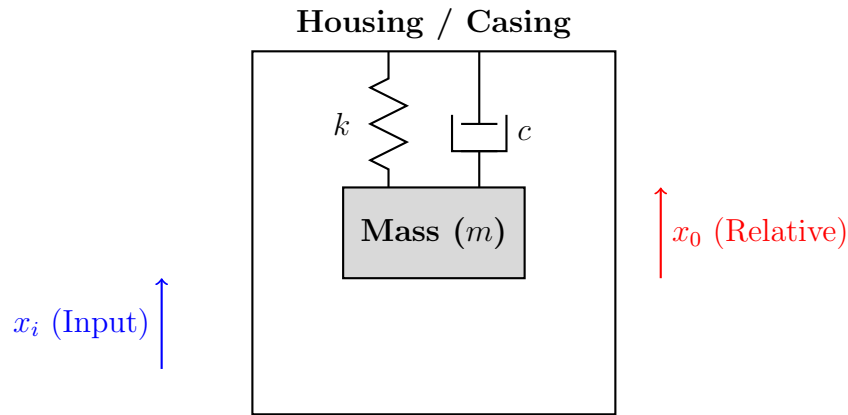
Conclusion: The differential transit time Δt is inversely proportional to the square of the speed of sound (c^2). Because c is a temperature-dependent variable, the statement is completely accurate.

Solution to Question 2.(b)

Problem Statement: The sensing region of a seismic type accelerometer behaves as a 2nd order mechanical system. The accelerometer has a damping ratio of 0.65. Its usable operating frequency range lies between 0 Hz to 2 kHz. A deviation of amplitude response of $\pm 5\%$ relative to its static value is tolerable. The amplitude of acceleration applied to the instrument is 200 m/s^2 . Calculate the value of the parameter 'damping' with proper unit for a seismic mass of 0.05 kg.

1. System Diagram

A seismic accelerometer consists of a proof mass (m), a spring (k), and a damper (c) housed inside a casing.



2. Given Data

- Damping ratio (ζ): 0.65
- Usable frequency range: $0 \leq f \leq 2000 \text{ Hz}$ (Max usable frequency $f_{max} = 2000 \text{ Hz}$)
- Tolerable amplitude deviation: $\pm 5\%$ relative to the static value.
- Seismic mass (m): 0.05 kg

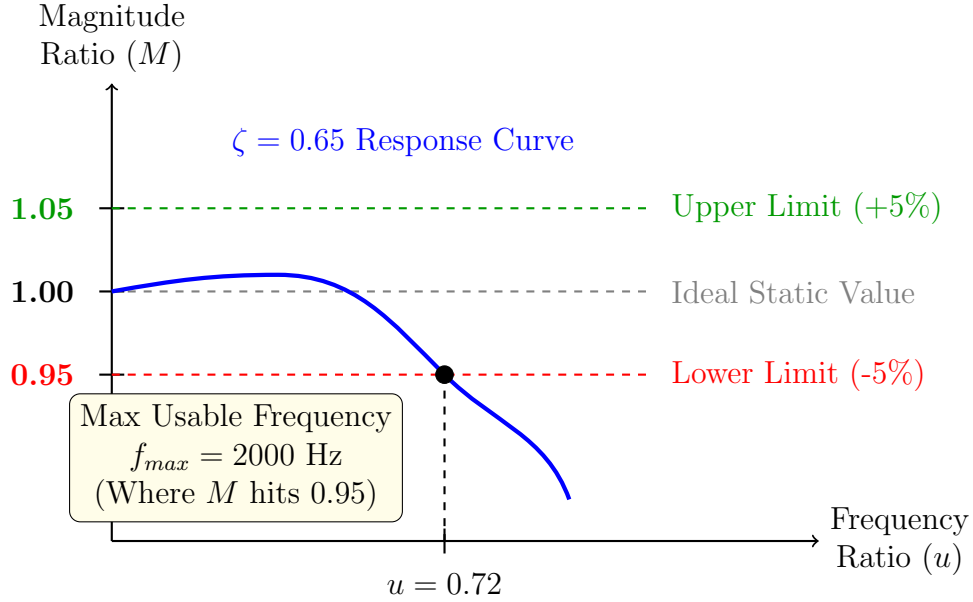
3. Amplitude Response Logic

For a seismic accelerometer, the normalized amplitude response (magnitude ratio M) is:

$$M = \frac{1}{\sqrt{(1 - u^2)^2 + (2\zeta u)^2}} \quad (24)$$

Where $u = \frac{f}{f_n}$ is the frequency ratio.

A tolerance of $\pm 5\%$ means the usable magnitude ratio must remain within $0.95 \leq M \leq 1.05$. In the visually magnified graph below, the curve stays within the tolerance limits until it drops to exactly 0.95 at the maximum usable frequency.



4. Finding the Maximum Frequency Ratio (u)

Setting $M = 0.95$:

$$0.95 = \frac{1}{\sqrt{(1-u^2)^2 + (2(0.65)u)^2}} \quad (25)$$

$$(0.95)^2 = \frac{1}{(1-u^2)^2 + 1.69u^2} \quad (26)$$

$$(1-u^2)^2 + 1.69u^2 = \frac{1}{0.9025} \quad (27)$$

$$1 - 2u^2 + u^4 + 1.69u^2 = 1.10803 \quad (28)$$

$$u^4 - 0.31u^2 - 0.10803 = 0 \quad (29)$$

Let $x = u^2$:

$$x^2 - 0.31x - 0.10803 = 0 \quad (30)$$

Applying the quadratic formula:

$$x = \frac{0.31 + \sqrt{(-0.31)^2 - 4(1)(-0.10803)}}{2} = \frac{0.31 + \sqrt{0.52822}}{2} = 0.5184 \quad (31)$$

Since $x = u^2$:

$$u = \sqrt{0.5184} = 0.72 \quad (32)$$

5. Calculating Natural Frequency (ω_n)

Using $f_{max} = 2000$ Hz at $u = 0.72$:

$$u = \frac{f_{max}}{f_n} \implies 0.72 = \frac{2000}{f_n} \quad (33)$$

$$f_n = \frac{2000}{0.72} = 2777.78 \text{ Hz} \quad (34)$$

Angular natural frequency (ω_n):

$$\omega_n = 2\pi f_n = 2\pi(2777.78) \approx 17453.29 \text{ rad/s} \quad (35)$$

6. Calculating the Damping Parameter (c)

The damping coefficient (c) is:

$$c = 2\zeta m\omega_n \quad (36)$$

Substituting the known values:

$$c = 2 \times 0.65 \times 0.05 \text{ kg} \times 17453.29 \text{ rad/s} \quad (37)$$

$$c = 0.065 \times 17453.29 \quad (38)$$

$$c \approx 1134.46 \text{ N} \cdot \text{s/m} \quad (39)$$

Final Answer:

The value of the damping parameter is approximately **1134.46 N·s/m** (or kg/s).

Solution to Question 3.(a)

Problem Statement: “A logarithmic amplifier can be used for linearizing the output versus temperature characteristic for NTC thermistor based temperature measurement system.” – Justify or correct this statement. Give suitable figure and derivations. *Practical example of any opamp based log circuit is not required.*

1. Justification & Correction

The statement is **partially correct but requires a precise technical correction.**

An NTC (Negative Temperature Coefficient) thermistor exhibits a highly non-linear, exponential relationship between its resistance and temperature. While a logarithmic amplifier is indeed the standard analog method used to counter this exponential behavior, it does not linearize the output with respect to the absolute temperature (T) directly.

Correction: A logarithmic amplifier linearizes the output voltage with respect to the **inverse of the absolute temperature** ($1/T$).

2. Derivation of the Linearized Output

Step 1: The Thermistor Characteristic Equation

The resistance-temperature relationship of an NTC thermistor is commonly approximated by the β -parameter equation:

$$R_T = R_0 \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \right] \quad (40)$$

Where:

- R_T = Resistance of the thermistor at absolute temperature T (in Kelvin)
- R_0 = Nominal resistance at a reference temperature T_0 (usually 298.15 K)
- β = Material constant of the NTC thermistor (in Kelvin)

We can rearrange the constants into a single constant A :

$$A = R_0 \exp \left(-\frac{\beta}{T_0} \right) \quad (41)$$

Thus, the resistance equation simplifies to:

$$R_T = A \exp \left(\frac{\beta}{T} \right) \quad (42)$$

Step 2: Conversion to Voltage

To measure the change in resistance, the thermistor is typically driven by a **constant current source** (I_c). The voltage drop across the thermistor (V_{in}) is:

$$V_{in} = I_c \cdot R_T = I_c \cdot A \exp \left(\frac{\beta}{T} \right) \quad (43)$$

Step 3: Applying the Logarithmic Amplifier

The output voltage (V_{out}) of an ideal logarithmic amplifier is proportional to the natural logarithm of its input voltage:

$$V_{out} = K \ln(V_{in}) \quad (44)$$

Where K is the scaling constant (gain) of the log amplifier.

Substituting V_{in} into the log amplifier equation gives:

$$V_{out} = K \ln \left[I_c \cdot A \exp \left(\frac{\beta}{T} \right) \right] \quad (45)$$

$$V_{out} = K \left[\ln(I_c \cdot A) + \ln \left(\exp \left(\frac{\beta}{T} \right) \right) \right] \quad (46)$$

$$V_{out} = K \ln(I_c \cdot A) + K \left(\frac{\beta}{T} \right) \quad (47)$$

Let $C_1 = K\beta$ and $C_2 = K \ln(I_c \cdot A)$. Both C_1 and C_2 are constants. The final expression becomes:

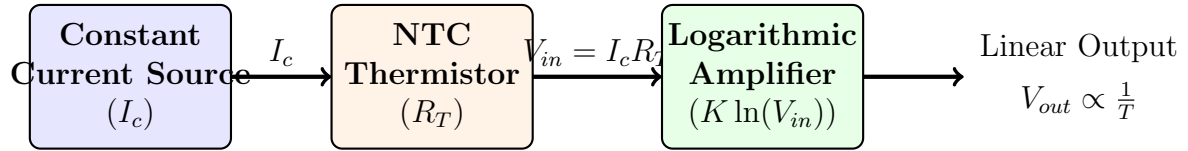
$$V_{out} = C_1 \left(\frac{1}{T} \right) + C_2 \quad (48)$$

Conclusion of Derivation:

The final equation is in the form of a straight line, $y = mx + c$, where $y = V_{out}$ and $x = 1/T$. Therefore, the output voltage is **strictly linear with respect to** $1/T$.

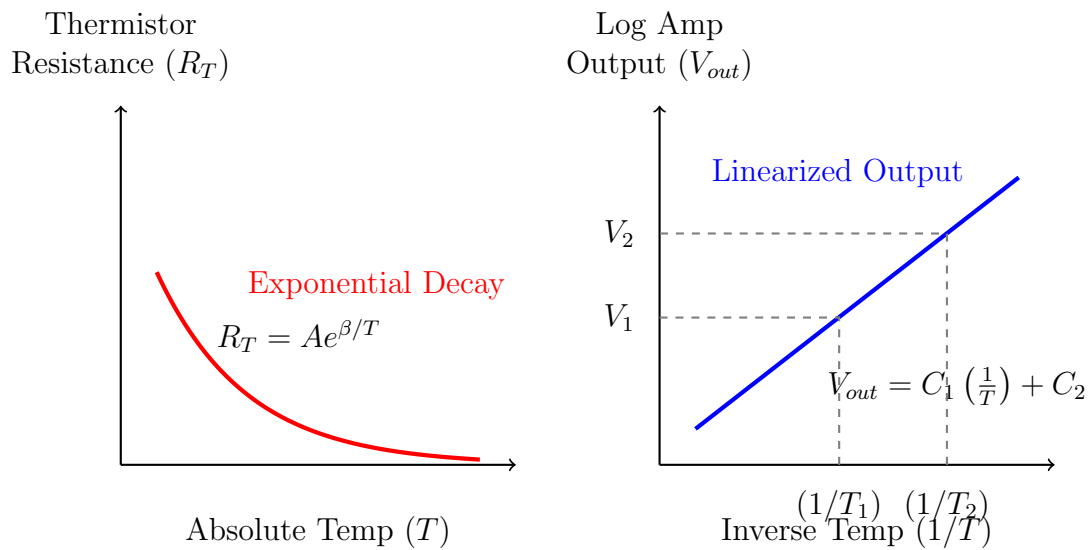
3. Block Diagram of the Measurement System

Below is the generalized block diagram showing how the thermistor's exponential characteristic is transformed into a linear output.



4. Graphical Representation

The graphs below clearly contrast the non-linear relationship of the raw sensor against the linearized output achieved via the log amplifier.

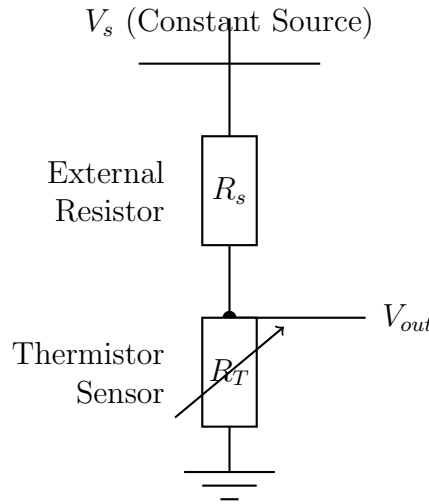


Solution to Question 3.(b)

Problem Statement: A temperature sensor has a nominal resistance of $1\text{ k}\Omega$ at 25°C . Its resistances at 40°C and 100°C are 1124Ω and 1706Ω respectively. It is to be used in an ohmmeter circuit, over 40°C to 100°C . How can you improve the linearity of the output voltage versus temperature characteristic by introducing an external resistance? Calculate the value of such resistance. Derive the expression used.

1. Proposed Circuit for Linearization

To improve the linearity of the output voltage versus temperature characteristic, the temperature sensor (R_T) is placed in a **Voltage Divider Circuit** in series with an external compensating resistor (R_s).



2. Mathematical Modeling

The sensor's resistance is modeled using a second-order polynomial:

$$R_T(T) = A + BT + CT^2$$

Using the three provided calibration points:

- At 25°C , $R_T = 1000\Omega \implies A + 25B + 625C = 1000$
- At 40°C , $R_T = 1124\Omega \implies A + 40B + 1600C = 1124$
- At 100°C , $R_T = 1706\Omega \implies A + 100B + 10000C = 1706$

Solving this system yields: $A = 812.45$, $B = 7.0244$, $C = 0.019111$.

For the specific operating range of 40°C to 100°C , we calculate the resistance at the limits and the exact midpoint (70°C):

- $T_1 = 40^\circ\text{C} \implies R_1 = 1124\Omega$

- $T_2 = 70^\circ\text{C} \implies R_2 = 812.45 + 70(7.0244) + 4900(0.019111) = 1397.8 \Omega$
- $T_3 = 100^\circ\text{C} \implies R_3 = 1706 \Omega$

3. Derivation of the External Resistance (R_s)

The output voltage is determined by the standard voltage divider formula:

$$V_{out} = V_s \left(\frac{R_T}{R_T + R_s} \right)$$

For perfect linearity between T_1 and T_3 , the change in voltage must be symmetrical around the midpoint:

$$\begin{aligned} V_{out}(T_2) - V_{out}(T_1) &= V_{out}(T_3) - V_{out}(T_2) \\ 2V_{out}(T_2) &= V_{out}(T_1) + V_{out}(T_3) \end{aligned}$$

Substitute the voltage divider equation and cancel out the source voltage V_s :

$$2 \left(\frac{R_2}{R_2 + R_s} \right) = \frac{R_1}{R_1 + R_s} + \frac{R_3}{R_3 + R_s}$$

Cross-multiplying and grouping the R_s terms yields the formula:

$$R_s = \frac{2R_1R_3 - R_1R_2 - R_2R_3}{2R_2 - R_1 - R_3}$$

4. Final Calculation

Substitute $R_1 = 1124 \Omega$, $R_2 = 1397.8 \Omega$, and $R_3 = 1706 \Omega$ into the derived equation:

$$\begin{aligned} \text{Numerator} &= 2(1124)(1706) - (1124)(1397.8) - (1397.8)(1706) \\ &= 3835088 - 1571127.2 - 2384646.8 = -120686 \end{aligned}$$

$$\begin{aligned} \text{Denominator} &= 2(1397.8) - 1124 - 1706 \\ &= 2795.6 - 2830 = -34.4 \end{aligned}$$

$$R_s = \frac{-120686}{-34.4} \approx 3508.31 \Omega$$

Conclusion:

Introducing an external series resistance of approximately **3.51 k Ω** will heavily mitigate the sensor's non-linearity, producing an optimized linear output voltage response across the 40°C to 100°C operating range.

Solution to Question 4

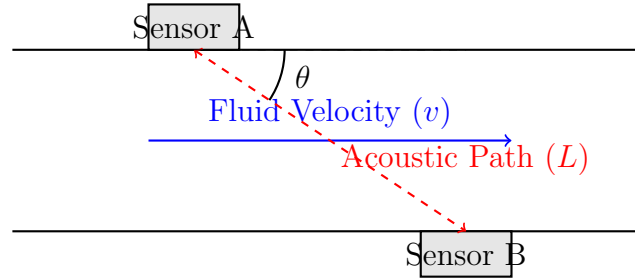
Problem Statement: Write short notes on the following: (a) Ultrasonic flowmeter of non-wetted sensor variety. (b) Level gauges for non-conducting liquids. (c) Synchronous demodulator for capacitive sensors.

4(a). Ultrasonic Flowmeter of Non-Wetted Sensor Variety

1. Working Principle

A non-wetted ultrasonic flowmeter is commonly known as a **clamp-on transit-time flowmeter**. The piezoelectric transducers are mounted directly onto the exterior of the pipe, meaning they never come into physical contact with the fluid. It operates by measuring the time of flight of an ultrasonic pulse traveling diagonally across the pipe, both with and against the direction of fluid flow.

2. System Diagram



3. Mathematical Derivation

Let c be the speed of sound in the fluid, v be the velocity of the fluid, L be the acoustic path length, and θ be the angle between the ultrasonic beam and the fluid flow.

The transit time for the pulse traveling **downstream** (from A to B, aided by flow) is:

$$t_{AB} = \frac{L}{c + v \cos \theta}$$

The transit time for the pulse traveling **upstream** (from B to A, opposed by flow) is:

$$t_{BA} = \frac{L}{c - v \cos \theta}$$

The differential transit time (Δt) is the difference between these two measurements:

$$\begin{aligned} \Delta t &= t_{BA} - t_{AB} = \frac{L}{c - v \cos \theta} - \frac{L}{c + v \cos \theta} \\ \Delta t &= \frac{L(c + v \cos \theta) - L(c - v \cos \theta)}{c^2 - v^2 \cos^2 \theta} = \frac{2Lv \cos \theta}{c^2 - v^2 \cos^2 \theta} \end{aligned}$$

Since $c^2 \gg v^2 \cos^2 \theta$ for liquids, the denominator simplifies to c^2 . The fluid velocity v can be isolated:

$$v \approx \frac{c^2 \Delta t}{2L \cos \theta}$$

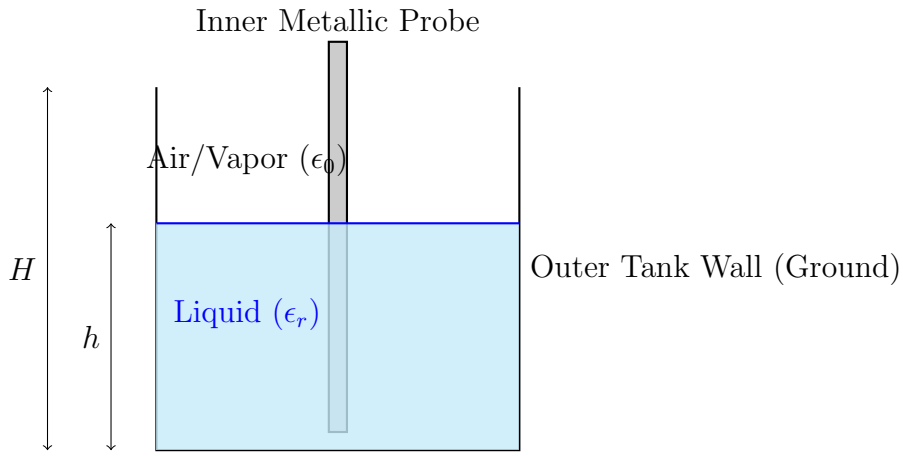
Advantages: Zero pressure drop, no corrosion/contamination since sensors are external to the pipe, and highly portable.

4(b). Level Gauges for Non-Conducting Liquids

1. Working Principle

For non-conducting (dielectric) liquids like oils or hydrocarbons, a **Capacitive Level Sensor** is universally employed. It acts as a large cylindrical capacitor. The metallic tank wall acts as the outer electrode, and a centrally inserted metallic rod acts as the inner electrode. The non-conducting liquid and the air/vapor above it act as two parallel dielectric mediums altering the total capacitance as the liquid level rises.

2. System Diagram



3. Mathematical Derivation

The system behaves as two capacitors connected in parallel: C_{liquid} (submerged portion) and C_{air} (unsubmerged portion). Let r_1 be the radius of the inner probe, r_2 be the radius of the tank, and ϵ_r be the relative permittivity of the liquid. The capacitance of a cylindrical capacitor is given by $C = \frac{2\pi\epsilon L}{\ln(r_2/r_1)}$.

The submerged liquid capacitance:

$$C_{liquid} = \frac{2\pi\epsilon_0\epsilon_r h}{\ln(r_2/r_1)}$$

The air/vapor capacitance above the liquid:

$$C_{air} = \frac{2\pi\epsilon_0(H - h)}{\ln(r_2/r_1)}$$

The total equivalent capacitance C_{total} is the sum of the two:

$$\begin{aligned} C_{total} &= C_{liquid} + C_{air} \\ C_{total} &= \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \left[\epsilon_r h + (H - h) \right] \\ C_{total} &= \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \left[h(\epsilon_r - 1) + H \right] \end{aligned}$$

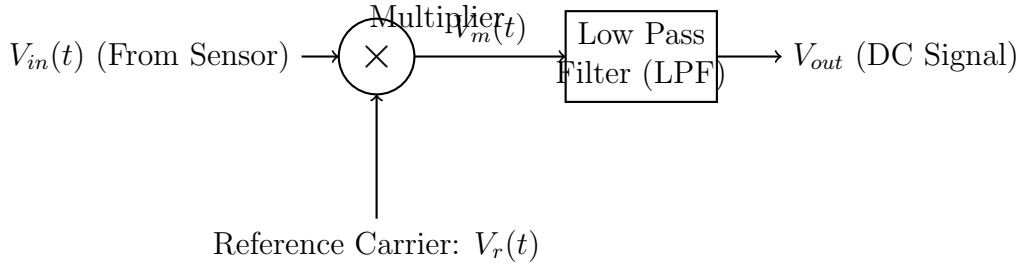
Because $\epsilon_r > 1$, the capacitance C_{total} increases linearly with the liquid height h .

4(c). Synchronous Demodulator for Capacitive Sensors

1. Working Principle

Capacitive sensors are often placed in an AC-excited Wheatstone bridge. The output is an amplitude-modulated (AM) suppressed-carrier signal where the amplitude represents the magnitude of the measured physical variable, and the phase (in-phase or 180° out-of-phase) represents the direction/sign. A **Phase-Sensitive Detector (Synchronous Demodulator)** is used to multiply the modulated sensor output by the original high-frequency reference carrier, followed by a low-pass filter (LPF) to extract the pure, noise-immune DC/low-frequency baseband signal.

2. System Diagram



3. Mathematical Derivation

Let the measurand (physical variable) be $x(t)$. The output from the AC bridge (input to demodulator) is:

$$V_{in}(t) = x(t) \cdot V_s \cos(\omega_c t)$$

The synchronous reference signal from the oscillator is:

$$V_r(t) = \cos(\omega_c t)$$

The multiplier mixes these two signals, producing $V_m(t)$:

$$\begin{aligned} V_m(t) &= V_{in}(t) \times V_r(t) \\ V_m(t) &= \left[x(t) V_s \cos(\omega_c t) \right] \times \cos(\omega_c t) \\ V_m(t) &= x(t) V_s \cos^2(\omega_c t) \end{aligned}$$

Using the trigonometric identity $\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$:

$$\begin{aligned} V_m(t) &= \frac{V_s \cdot x(t)}{2} \left[1 + \cos(2\omega_c t) \right] \\ V_m(t) &= \underbrace{\frac{V_s \cdot x(t)}{2}}_{\text{Baseband (Desired)}} + \underbrace{\frac{V_s \cdot x(t)}{2} \cos(2\omega_c t)}_{\text{High Frequency (Rejected)}} \end{aligned}$$

The signal passes through the Low Pass Filter (LPF) with a cutoff frequency well below $2\omega_c$, stripping away the high-frequency alternating component. The final extracted output is:

$$V_{out} = \frac{V_s}{2} \cdot x(t)$$

Conclusion: The output is perfectly proportional to the physical variable $x(t)$, completely free of external out-of-band noise, and correctly indicates polarity based on the relative phase.

EXAMINATION SOLUTIONS: PART I

Rigorous Justifications, Derivations, and Equivalent Circuits

Question 5(a): Astable Multivibrator Linearization

Problem Statement

"In an astable multivibrator circuit used for variable air-gap capacitive sensors, the deviation of the output frequency from its nominal value is linearly related with the displacement being measured". Justify or correct the statement with necessary illustration and derivation.

Verdict

The statement is **CORRECT**.

The Mathematical Proof: A variable air-gap capacitive sensor suffers from severe hyperbolic non-linearity when measuring displacement (x). Its capacitance is given by:

$$C(x) = \frac{\epsilon A}{d \pm x} \quad (1)$$

where d is the nominal resting gap, A is the area, and ϵ is the permittivity. Let us consider the case where the gap increases ($d + x$).

If this non-linear capacitor is placed in an Astable Multivibrator (e.g., a 555 Timer circuit) to act as the timing capacitor, the time period of oscillation (T) is directly proportional to the capacitance:

$$T = K_R \cdot C(x) \quad \text{where } K_R = \ln(2) \cdot (R_A + 2R_B) \quad (2)$$

The output frequency (f) of the multivibrator is the mathematical inverse of the time period:

$$f = \frac{1}{T} = \frac{1}{K_R \cdot C(x)} \quad (3)$$

Substitute the variable gap capacitance $C(x)$ directly into the frequency equation:

$$f = \frac{1}{K_R \left(\frac{\epsilon A}{d+x} \right)} = \frac{d+x}{K_R \epsilon A} \quad (4)$$

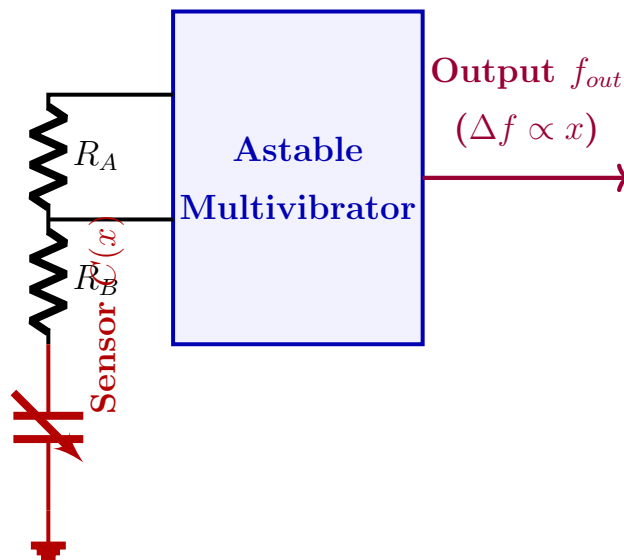
We can split this fraction into two distinct components:

$$f = \left(\frac{d}{K_R \epsilon A} \right) + \left(\frac{1}{K_R \epsilon A} \right) x \quad (5)$$

The first term is the nominal resting frequency (f_0) when $x = 0$. Therefore:

$$f = f_0 + K \cdot x \implies \Delta f = f - f_0 = K \cdot x \quad (6)$$

Conclusion: The deviation in frequency (Δf) is strictly equal to a constant (K) multiplied by the displacement (x). The hyperbolic $1/(d+x)$ non-linearity is mathematically annihilated, proving the statement is perfectly correct.



Question 5(b): Transformer Ratio Bridge

Problem Statement

Show that the performance of a transformer double ratio bridge for capacitance measurement is inherently free from the effect of parasitic capacitances.

The Conceptual and Mathematical Proof: When measuring small capacitances (< 100 pF), standard AC Wheatstone bridges fail catastrophically because parasitic stray capacitances (C_{p1}, C_{p2}) between the measurement nodes and the grounded metallic chassis completely dominate the balance equations.

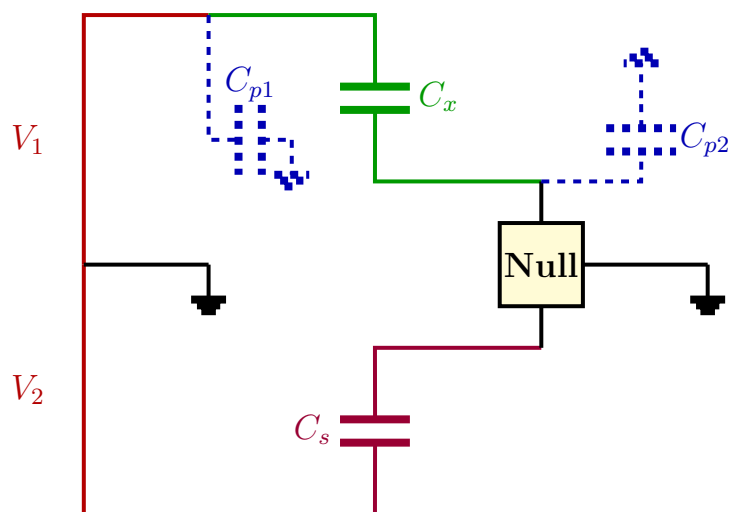
The Transformer Ratio Bridge solves this inherently through two physical properties:

1. Low Source Impedance: The transformer acts as a near-ideal voltage source with practically zero internal leakage impedance. **2. The Grounded Null Node:** The detector is connected directly to ground.

Let C_{p1} be the stray capacitance across the transformer secondary, and C_{p2} be the stray capacitance at the central detector node.

- **Effect of C_{p1} :** This stray capacitor is placed directly across the rigid output voltage of the transformer winding (V_1). Because the transformer has infinitesimally low internal impedance, it easily supplies the required reactive charging current to C_{p1} without allowing the node voltage V_1 to sag. The ratio $V_1/V_2 = N_1/N_2$ remains absolutely undisturbed.
- **Effect of C_{p2} :** At the exact null balance point, the voltage at the detector node is exactly zero (0V relative to ground). Since the potential difference across C_{p2} is strictly zero, no parasitic current can physically flow through it ($I = V/Z = 0$). It mathematically vanishes from the KCL balancing equation.

Conclusion: Because C_{p1} cannot shift the excitation voltage, and C_{p2} conducts zero current at null, the bridge balancing equation $C_x = C_s(N_2/N_1)$ is inherently immune to all parasitic effects without requiring complex Wagner Earth circuits.



Question 5(c): LVDT Output Phase Shift

Problem Statement

"In an LVDT, the differential output voltage is in phase with the input voltage." - Comment whether the statement is true or false. Justify in favour of your argument.

Verdict and Correction

Verdict: The statement is **FALSE**.

Correction: The differential output voltage inherently **leads** the primary input excitation voltage by an acute phase angle due to the internal resistance and inductance of the primary coil.

The Mathematical Proof: Let the primary excitation voltage be V_{in} . The primary coil is not a pure inductor; it possesses an internal resistance (R_p) and inductance (L_p). By Kirchhoff's Voltage Law, the primary current I_p is:

$$I_p = \frac{V_{in}}{R_p + j\omega L_p} \quad (7)$$

This current produces an alternating magnetic flux. By Faraday's Law, the induced EMF in the secondary coils depends on the rate of change of flux ($d\phi/dt$), which mathematically translates to multiplication by $j\omega$ (a strict $+90^\circ$ phase shift) and the mutual inductances (M_1, M_2).

$$E_{s1} = j\omega M_1 I_p \quad \text{and} \quad E_{s2} = j\omega M_2 I_p \quad (8)$$

The differential output voltage of the LVDT is $V_{out} = E_{s1} - E_{s2}$.

$$V_{out} = j\omega(M_1 - M_2)I_p \quad (9)$$

Substitute the primary current I_p into the output equation:

$$V_{out} = j\omega(M_1 - M_2) \left[\frac{V_{in}}{R_p + j\omega L_p} \right] \quad (10)$$

To find the phase angle, we convert to polar form. The numerator $j\omega$ contributes exactly $+90^\circ$. The denominator ($R_p + j\omega L_p$) contributes an angle of $\phi_p = \tan^{-1}(\omega L_p / R_p)$.

$$V_{out} = \left| \frac{\omega(M_1 - M_2)}{\sqrt{R_p^2 + \omega^2 L_p^2}} \right| \angle(90^\circ - \phi_p) \cdot V_{in} \quad (11)$$

Conclusion: Because the primary coil has a finite real resistance R_p , the angle ϕ_p is strictly less than 90° . Therefore, the net phase shift $\theta = (90^\circ - \phi_p)$ is a **positive acute angle**. The output voltage V_{out} physically leads the input voltage V_{in} , proving the statement false. (This requires an R-C low-pass filter to correct).

